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Salience, Myopia, and Complex Dynamic Incentives: Evidence from Medicare Part D

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1 Big picture

- **Question.** Do consumers respond to complex dynamic incentives in health insurance as forward-looking optimizers, or do they react mainly to the price that is salient at the current point of purchase?
- **Setting.** The paper studies Medicare Part D prescription drug insurance in 2008, focusing on enrollees who are close to the Part D coverage gap, the “doughnut hole.”
- **Policy feature.** The standard Part D contract is nonlinear: enrollees face partial insurance in the initial coverage region, then much higher out-of-pocket prices in the coverage gap, and then catastrophic coverage after very high out-of-pocket spending.
- **Neoclassical prediction.** Enrollees who arrive near the gap early in the year should already expect to enter the gap. A forward-looking consumer should incorporate the future gap price before the gap starts, so spending should not drop sharply exactly when entering the gap.
- **Reduced-form evidence.** Spending is flat before the gap and drops sharply once the gap begins. Total weekly spending falls by about \$17–18, the number of prescriptions falls, and the drop is strongest for branded and expensive drugs.
- **Structural result.** The structural estimates favor a price-salience model in which future prices are not salient. The estimated salience parameter is at the boundary: future prices are effectively ignored.
- **Model comparison.** The price-salience model fits better than quasi-hyperbolic discounting. It captures the sharp discontinuous spending drop at the gap more closely.
- **Counterfactual result.** If consumers behaved according to the near-neoclassical benchmark, weekly enrollee spending would fall by 31% and total drug spending by 15% relative to the estimated baseline.
- **Policy result.** Eliminating the doughnut hole increases total spending by about 10% and insurer spending by 27%; filling the gap only for generics gives a smaller insurer-cost increase.
- **Economic takeaway.** Nonlinear insurance contracts can create behavioral hazard: consumers respond strongly to current salient prices and weakly to future prices embedded in the contract.

2 Incremental contribution of this paper

- **Beyond plan-choice mistakes.** Earlier Part D work emphasizes that consumers choose suboptimal plans. This paper studies a different margin: whether consumers dynamically optimize *within* a plan when making weekly prescription purchases.
- **Beyond standard moral hazard.** Standard health-insurance work studies responses to current cost sharing. This paper separates conventional insurance effects from *behavioral hazard*, meaning mistakes caused by limited salience or myopia in a nonlinear contract.
- **A clean within-person test.** Rather than comparing different people across different plans, the reduced-form design follows the same enrollee as she moves from just before the gap to just inside the gap.
- **A testable prediction of dynamic optimization.** The paper shows that for enrollees who expect to reach the gap with high probability, the benchmark model predicts no sharp spending decline at the gap. This makes the gap-crossing discontinuity informative about behavior.
- **Separating salience from discounting.** The structural model compares geometric discounting, quasi-hyperbolic discounting, and price salience. The observed pattern—flat pre-gap spending plus a sharp gap drop—is especially informative for distinguishing low discounting from low salience.
- **Policy relevance.** The paper quantifies how much spending and welfare change under counterfactual redesigns of the doughnut hole, including full elimination and generic-only coverage.

Interpretation: The incremental contribution is not just that consumers react to the doughnut hole. The deeper contribution is to show that the *timing and shape* of the response identifies limited future price salience in a dynamic nonlinear pricing problem.

3 Data and measurement

3.1 Medicare Part D contract

- Medicare Part D provides prescription drug coverage through private plans.
- In 2008, the standard contract had three main regions: initial coverage, coverage gap, and catastrophic coverage.
- The coverage gap started at \$2,510 in total spending.
- The catastrophic region started at \$4,050 in out-of-pocket spending, corresponding to roughly \$5,932.50 in mean total spending.
- Inside the gap, enrollees paid the full price of drugs until reaching the catastrophic threshold.

Interpretation: The nonlinear schedule creates a dynamic pricing problem. A purchase today can move the consumer closer to a future region with much higher prices.

3.2 Main sample

- The analysis uses 2008 claims data from six employer-sponsored Part D plans.
- The base estimation sample focuses on enrollees who start a week between March 30 and July 20 with cumulative total spending in the range \$2,000–\$2,510.
- These enrollees are close enough to the gap for the current decision to matter but early enough in the year that they should expect to enter the gap with high probability.
- Across the base plans, 94–97% of the estimation-sample enrollees hit the coverage gap during the year.
- Plan F is used as a falsification plan because its coverage gap starts at \$4,000, not \$2,510.

Interpretation: The sample choice is crucial. The authors deliberately study people for whom a rational forward-looking model says the gap should already be priced in before the threshold is crossed.

3.3 Health shocks and drug choice

- Each drug is assigned to a health-shock type based on clinical function.
- The most common health shock types include cholesterol-lowering drugs, renin-angiotensin blockers, beta-blockers, opioids, antidepressants, and diuretics.
- The structural model treats a weekly drug decision as a discrete-choice problem: conditional on a health shock, the enrollee chooses one of the relevant drugs or the outside option.
- Prices differ across drugs, branded status, and coverage-region position, providing identifying variation for price responsiveness and behavioral parameters.

Interpretation: Classifying drugs by health-shock type matters because it prevents the model from comparing clinically unrelated drugs as if they were close substitutes.

3.4 Key empirical outcomes

- Weekly total spending.
- Mean price per prescription.
- Number of prescriptions.
- Number of branded prescriptions and generic prescriptions.
- Number of expensive, medium-price, and inexpensive prescriptions.
- Counterfactual enrollee spending, insurer spending, total spending, and welfare.

Interpretation: The paper does not measure only whether people stop taking drugs. It also studies substitution toward cheaper, generic, and less expensive drugs, which is important for distinguishing price response from health-shock timing.

4 Model and empirical specifications

4.1 Effective price in the fully salient benchmark

Let m be the dollars remaining until the doughnut hole, p_{hj} be the full price of drug j for health shock h , and oop_{hj} be the out-of-pocket price in the initial coverage region. Under full salience, the perceived effective price is

$$p^{\text{eff}}(m, p_{hj}, \text{oop}_{hj}) = \begin{cases} p_{hj}, & 0 \leq m < \text{oop}_{hj}, \\ \text{oop}_{hj} + p_{hj} - m, & \text{oop}_{hj} \leq m < p_{hj}, \\ \text{oop}_{hj}, & p_{hj} \leq m. \end{cases} \quad (1)$$

Interpretation: This function captures the Part D transition rule. If the current purchase crosses the gap threshold, the enrollee pays the ordinary copay up to the threshold and then the full price inside the gap.

4.2 Flow utility from a drug choice

For drug j treating health shock h , the perceived flow utility is

$$u_{hj} = \phi_{hj} - \alpha(p^{\text{eff}}(m, p_{hj}, \text{oop}_{hj})) - c_h + \varepsilon_{hj}, \quad (2)$$

where ϕ_{hj} is drug fixed utility, $\alpha(\cdot)$ is price disutility, c_h is the health cost common across treatment options, and ε_{hj} is an idiosyncratic type-I extreme-value term.

Interpretation: Conditional on a health shock, the model is a multinomial logit choice among drugs and the outside option, embedded inside a dynamic problem.

4.3 Benchmark dynamic optimization

In the benchmark model, consumers use geometric discounting and fully internalize how current purchases affect future coverage-region prices:

$$V(m, n) = \max_j \{ \phi_{hj} - \alpha(p^{\text{eff}}(m, p_{hj}, \text{oop}_{hj})) - c_h + \delta V(\max\{m - p_{hj}, 0\}, n + 1) + \varepsilon_{hj} \}. \quad (3)$$

Interpretation: If δ is close to one and the consumer knows she will enter the gap, she should act before the gap as if the future gap price already matters. Therefore a sharp spending drop exactly at the gap is not predicted.

4.4 Quasi-hyperbolic discounting

The quasi-hyperbolic model changes the current-versus-future tradeoff by weighting continuation value with β :

$$\max_j \left\{ \phi_{hj} - \alpha(p^{\text{eff}}(m, p_{hj}, \text{oop}_{hj})) - c_h + \beta \delta V(\max\{m - p_{hj}, 0\}, n + 1) + \varepsilon_{hj} \right\}. \quad (4)$$

The paper considers both naive and sophisticated quasi-hyperbolic consumers.

Interpretation: Low β creates present bias. However, the pattern of spending away from the gap differs from pure low salience, which lets the authors compare the models.

4.5 Price salience

In the price-salience model, future prices are only partially incorporated in current choices. The behavioral parameter σ indexes how much future price consequences are salient. The estimated boundary value $\sigma = 0$ means future prices are not salient.

Interpretation: Salience is not the same as low patience. A low-patience consumer may still foresee the future but discount it. A low-salience consumer may simply fail to incorporate future prices until they become current.

4.6 Reduced-form discontinuity regressions

The reduced-form evidence compares weekly outcomes just before and just after the doughnut-hole threshold:

$$Y_{it} = \alpha_i + \gamma \mathbf{1}\{\text{beginning-of-week spending in \$2,510--\$2,999}\} + \varepsilon_{it}. \quad (5)$$

A second specification adds a pre-gap slope indicator for spending in the \$2,200–\$2,399 region.

Interpretation: The key coefficient is γ . The benchmark predicts no large discontinuity for the selected sample. A large negative γ indicates behavior changes when the gap price becomes current and salient.

4.7 Structural likelihood

The structural likelihood integrates over unobserved outside-option health shocks. If health shocks were fully observed, the log likelihood would be a product of health-shock arrival probabilities, type probabilities, and drug-choice probabilities. Because the outside option is unobserved, the likelihood sums over configurations of unobserved outside-option choices.

Interpretation: This is why the model is more complex than a static logit. The data show purchased drugs, not all health shocks that might have generated no purchase.

5 Identification and empirical design

5.1 Why the discontinuity is informative

- The same enrollee is compared before and after entering the coverage gap.
- The sample is restricted to enrollees who should expect to hit the gap with high probability.
- The benchmark model with high δ predicts little change exactly at gap entry.
- A low geometric discount factor predicts a gradual decline before the gap, not a flat pre-gap region and sharp gap drop.
- Price salience predicts flat pre-gap spending and a drop only when the high price becomes salient.

Interpretation: The design uses the shape of the spending path, not just the existence of a response, to distinguish behavioral models.

5.2 Main identifying assumptions

- Enrollees in the sample expect to enter the doughnut hole even if they change one purchase.
- They do not expect to reach catastrophic coverage.
- The timing of crossing \$2,510 is not entirely caused by a medical event that also changes future demand.
- Drug-price heterogeneity within health-shock types is sufficient to identify price elasticities and behavioral parameters.
- Plan F, where the gap begins at \$4,000, serves as a falsification check for the \$2,510 discontinuity.

Interpretation: The paper is strongest where the near-gap sample is plausibly comparable on both sides of the threshold and where the observed pattern is not replicated in the falsification plan.

5.3 Threats and responses

- **Medical shocks.** A sudden health improvement exactly at the gap could mimic lower spending. The falsification and flat pre-gap spending help address this.
- **Drug stockpiling.** If patients stockpile before the gap, spending could fall afterward. The authors' weekly design and pre-gap slope tests reduce this concern, though it cannot be eliminated completely.
- **Plan selection.** The main discontinuity test is within plan and within enrollee, limiting cross-plan selection concerns.
- **Physician agency.** Doctors may influence prescriptions, but the behavioral response still shows up in the realized drug purchase pattern.

Interpretation: The paper is not claiming a randomized experiment. Its credibility comes from a sharp, model-specific prediction combined with within-person movement across the nonlinear contract.

6 Tables and figures

This section groups adjacent visuals to save space. The tables and figures are directly cropped from the original paper.

6.1 Figure 1 and Table 1: Part D nonlinear contract and plan sample

Read:

- Figure 1 shows the 2008 standard Part D benefit: initial coverage, then the coverage gap starting at \$2,510 in total spending, then catastrophic coverage at high out-of-pocket spending.
- Table 1 shows that the main plans A–E share the same \$2,510 doughnut-hole threshold, while Plan F starts the gap at \$4,000 and is reserved for falsification.
- In the estimation sample, 94–97% of enrollees in base plans hit the \$2,510 threshold, supporting the assumption that they should expect to enter the gap.
- Mean total spending in the estimation sample is around \$3,867–\$4,284 across base plans, and mean age is in the early-to-mid 70s.

Economic message: The nonlinear contract creates the dynamic incentive problem. The sample is designed so that forward-looking enrollees should already understand that the gap is coming.

Additional interpretation: Table 1 is more than descriptive. It validates the paper’s central sample restriction: these are not low-spending consumers who might or might not hit the gap; they are near-gap consumers for whom the future gap should be relevant under the benchmark.

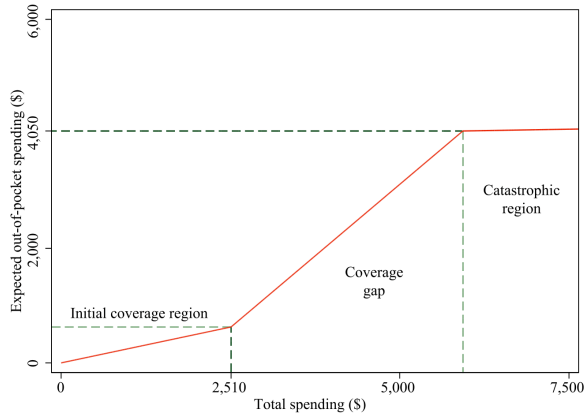


FIGURE 1
Coverage by region in 2008 with standard Medicare Part D plans

(a) Figure 1: coverage regions

	A	B	Plan C	D	E	F
Employer	1	1	1	2	2	3
% of employees from employer	26	45	9	79	21	46
Deductible (\$)	275	100	100	0	200	0
Doughnut hole start (total \$)	2,510	2,510	2,510	2,510	2,510	4,000
Catastrophic start (out-of-pocket \$)	4,050	4,050	4,050	4,010	4,010	4,050
Total enrollment	7,541	12,858	2,431	4,062	1,058	35,395
% hitting \$2,510	20	13	16	16	13	20
% hitting catastrophic coverage	2	1	1	1	1	0
Estimation sample:						
Enrollment	620	644	126	304	49	2,981
% hitting \$2,510	96	94	95	97	94	97
% hitting catastrophic	11	6	9	10	12	0
Mean total spending (\$)	4,284	3,867	4,009	4,246	3,974	4,072
Mean out-of-pocket (\$)	2,373	2,010	2,125	2,045	2,071	1,026
Mean age	74	73	73	75	75	78
Percent female	62	58	53	62	59	64
Mean ACG score	1.04	1.17	1.18	0.91	1.07	0.67

Note: Plan A provides generic coverage in deductible region; Plan F used for falsification exercise only and provides generic coverage in doughnut hole.

(b) Table 1: plan characteristics

6.2 Figures 2 and 3: model predictions before the gap

Read:

- Figure 2 shows simulations of the geometric model across discount factors. With $\delta = 0.999$, spending is essentially flat before the gap and inside the gap.
- With very low discounting, spending falls earlier and more continuously, rather than dropping only at the gap.

- Figure 3 compares behavioral models. Present bias and price salience can generate a sharp drop at the gap, but they have different predictions near the threshold.
- The salience model implies that consumers may not react before the high price becomes current, while quasi-hyperbolic consumers may react earlier if the current purchase itself moves them toward the gap.

Economic message: The simulation figures turn the theory into visual identifying predictions. The actual data should look flat before the gap if the benchmark is right, gradually falling if low geometric discounting is right, and sharply discontinuous if salience is limited.

Additional interpretation: The key contribution of these figures is to show that “myopia” is not a single empirical pattern. Low discount factors, present bias, and low salience can produce different pre-gap slopes and different discontinuities.

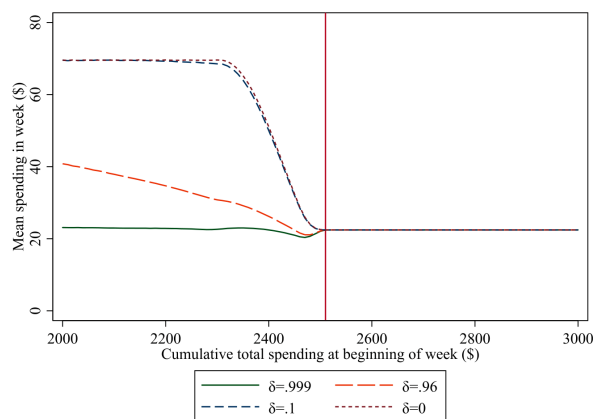


FIGURE 2
Simulated drug spending for the geometric model across discount factors

(a) Figure 2: geometric discounting

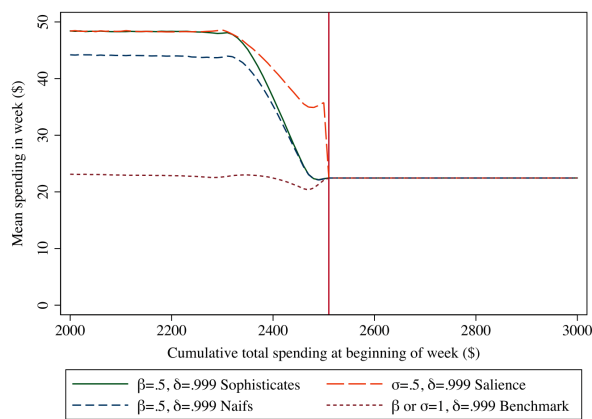


FIGURE 3
Simulated drug spending for different behavioural models

(b) Figure 3: behavioral models

6.3 Table 2 and Figure 4: health shocks and reduced-form gap response

Read:

- Table 2 shows the most common health shock types. Cholesterol-lowering prescriptions account for 9.4% of observations, followed by renin-angiotensin blockers at 7.9%.
- The most common prescriptions are ordinary chronic-care drugs such as simvastatin, lisinopril, metoprolol, and hydrocodone.
- Figure 4 shows mean weekly spending around the \$2,510 gap threshold. Spending is roughly flat before the gap and then drops once the threshold is crossed.
- The smoothed curve falls from the low \$60 range before the gap toward roughly the high \$40 range inside the gap.

Economic message: The behavioral response is not confined to rare or unusual drugs. It occurs in a setting dominated by common chronic-medication categories.

Additional interpretation: Figure 4 is the paper’s central reduced-form evidence. The shape rules out a simple low-discount-factor story: low discounting would imply a slope before the gap, but the data are mostly flat before the gap and lower only after entry.

TABLE 2
Most common health shock types in base estimation sample

Health shock type	Number Rx	% of obs.	Most common Rx
Cholesterol-Lowering	2,143	9.4	Simvastatin
Renin-Angiotensin System Blocker	1,814	7.9	Lisinopril
Beta-Blocker	1,259	5.5	Metoprolol
Opioid	1,200	5.2	Hydrocodon
Antidepressant	1,190	5.2	Sertraline
Diuretic	1,183	5.2	Furosemide
Calcium Channel Blocker	933	4.1	Amlodipine
Insulin Sensitizer	792	3.5	Metformin
Gastroesophageal Reflux & Peptic Ulcer	778	3.4	Omeprazole
Hypothyroidism	774	3.4	Levothyroxine

(a) Table 2: health shock types

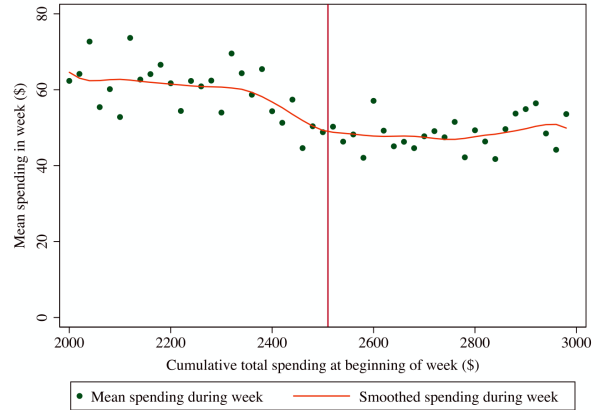


FIGURE 4
Spending near coverage gap for base estimation sample

(b) Figure 4: spending near the gap

6.4 Tables 3 and 4: discontinuity estimates

Read:

- Table 3 estimates that weekly spending drops by \$17.46 after entering the \$2,510–\$2,999 region, from a pre-gap mean of \$61.97.
- Mean price per prescription falls by \$9.77; the number of prescriptions falls by 0.18 from a pre-gap mean of 0.84.
- Branded prescriptions fall by 0.08, generic prescriptions by 0.10, expensive prescriptions by 0.04, and medium prescriptions by 0.06.
- Table 4 adds an indicator for the \$2,200–\$2,399 pre-gap region. The pre-gap spending slope is small and statistically insignificant, while the gap coefficient remains large and negative.
- The number of generic prescriptions shows some pre-gap movement, but total spending and expensive drug use do not show a large pre-gap decline.

Economic message: The spending response is both extensive and compositional. Consumers buy fewer prescriptions and shift away from higher-price drugs when the full price becomes salient.

Additional interpretation: The comparison between Tables 3 and 4 is important. Table 3 shows a big gap drop; Table 4 shows that the drop is not simply the continuation of a downward pre-gap trend. This strengthens the salience interpretation.

TABLE 3
Behaviour for sample arriving near coverage gap

Dependent variable:	Mean value before \$2,400	Beginning of week spending in: \$2,510–2,999	N
Mean spending in week	61.97	–17.46** (1.38)	28,542
Mean price per Rx	79.47	–9.77** (1.37)	10,840
Number of Rxs	0.84	–0.18** (0.02)	28,542
Number of branded Rxs	0.30	–0.08** (0.01)	28,542
Number of generic Rxs	0.54	–0.10** (0.01)	28,542
Expensive Rxs	0.12	–0.04** (0.00)	28,542
Medium Rxs	0.23	–0.06** (0.01)	28,542
Inexpensive Rxs	1.10	–0.01 (0.01)	28,542

Notes: Standard errors are in parentheses. *** denotes significance at the 1% level and ** at the 5% level. Each row represents one regression. All regressions also include enrollee fixed effects and an indicator for beginning-of-week spending between \$2,400 and \$2,509, and cluster standard errors at the enrollee level. An observation is an enrollee/week for an enrollee in the base estimation sample and beginning-of-week spending $\geq$ \$2,000 and $<$ \$3,000. Inexpensive Rxs are less than \$50 and expensive ones are \$150 or more.

(a) Table 3: behavior near the gap

TABLE 4
Behaviour near coverage gap with variation in pre-coverage gap region

Dependent variable:	Mean value before \$2,400	Beginning of week spending in: \$2,510–2,999	Beginning of week spending in: \$2,200–2,399	N
Mean spending in week	61.97	–17.79** (1.76)	–0.68 (2.25)	28,542
Mean price per Rx	79.47	–8.97** (1.72)	1.64 (2.13)	10,840
Number of Rxs	0.84	–0.20** (0.02)	–0.03 (0.03)	28,542
Number of branded Rxs	0.30	–0.08** (0.01)	0.01 (0.01)	28,542
Number of generic Rxs	0.54	–0.12** (0.02)	–0.04* (0.02)	28,542
Expensive Rxs	0.12	–0.04** (0.01)	–0.00 (0.01)	28,542
Medium Rxs	0.23	–0.06** (0.01)	0.00 (0.01)	28,542
Inexpensive Rxs	1.10	–0.02* (0.01)	–0.01 (0.02)	28,542

Notes: Standard errors in parentheses. *** denotes significance at the 1% level and ** at the 5% level. Each row represents one regression. All regressions also include enrollee fixed effects and an indicator for beginning-of-week spending between \$2,400 and \$2,509, and cluster standard errors at the enrollee level. An observation is an enrollee/week for an enrollee in the base estimation sample and beginning-of-week spending $\geq$ \$2,000 and $<$ \$3,000. Inexpensive Rxs are less than \$50 and expensive ones are \$150 or more.

(b) Table 4: pre-gap variation

6.5 Table 5 and Figure 5: structural estimation and model fit

Read:

- Table 5 estimates negative price coefficients across price ranges, confirming that consumers respond to out-of-pocket prices.
- The behavioral parameter is estimated at zero in all three behavioral specifications: $\beta = 0$ in quasi-hyperbolic models and $\sigma = 0$ in the salience model.
- The price-salience model has a better log likelihood than the quasi-hyperbolic models: -95,456.7 versus -95,594.6.
- The likelihood-ratio-style tests reject positive behavioral parameters and positive discount factors in the tested grids.
- Figure 5 shows that the salience model better matches the sharp drop at the gap. The quasi-hyperbolic model predicts too early a decline.

Economic message: Consumers are price sensitive, but the future part of the nonlinear price schedule is not salient. This is why the gap matters when it arrives, not before.

Additional interpretation: Table 5 and Figure 5 together distinguish two ideas that often get conflated. A consumer can be very price sensitive to current prices and still fail to incorporate future prices. The estimates say the problem is not indifference to price; it is lack of future-price salience.

TABLE 5
Main results of structural estimation

Model:	Quasi-hyperbolic discounting: naïfs	Quasi-hyperbolic discounting: sophisticates	Price salience
Price spline < \$20	-0.116** (0.006)	-0.116** (0.006)	-0.148** (0.007)
Price spline ∈ [\$20, \$50)	-0.012** (0.002)	-0.012** (0.002)	-0.014** (0.002)
Price spline ∈ [\$50, \$150)	-0.013** (0.001)	-0.013** (0.001)	-0.018** (0.001)
Price spline ≥ \$150	-0.006** (0.001)	-0.006** (0.001)	-0.003* (0.001)
Behavioural parameter: β or σ	0 (-)	0 (-)	0 (-)
Discount factor: δ	-	-	-
log L	-95,594.6	-95,594.6	-95,456.7
log L β or $\sigma = 0.1, \delta = 0.1$	-95,602.6	-95,602.4	-95,460.4
p-value for LM test	0.00	0.00	0.00
log L β or $\sigma = 0.1, \delta = 0.4$	-95,604.8	-95,604.5	-95,462.0
p-value for LM test	0.00	0.00	0.00
log L at β or $\sigma = 0.1, \delta = 0.995$	-95,619.6	-95,615.7	-95,471.6
p-value for LM test	0.00	0.00	0.00
log L at β or $\sigma = 0.3, \delta = 0.995$	-95,672.5	-95,663.8	-95,532.9
p-value for LM test	0.00	0.00	0.00
Number of health shock types H		60	
Number of drug fixed effects ϕ		245	
N		18,897	

Notes: Standard errors reported in parentheses are calculated using standard outer product approximations, treating β , σ and δ as fixed. *** denotes significance at the 1% level and ** at the 5% level. An observation is an enrollee/week for an enrollee in the base estimation sample and beginning-of-week spending ≥ \$2,000 and < \$3,000, with a middle ACG score. Each column displays the results from the maximum likelihood estimation for one model. Reported price coefficients are $-\alpha(-)$; all prices affect utility negatively. All specifications also include fixed effects ϕ for each drug. LM tests are for the restrictions on β or σ and δ .

(a) Table 5: structural estimates

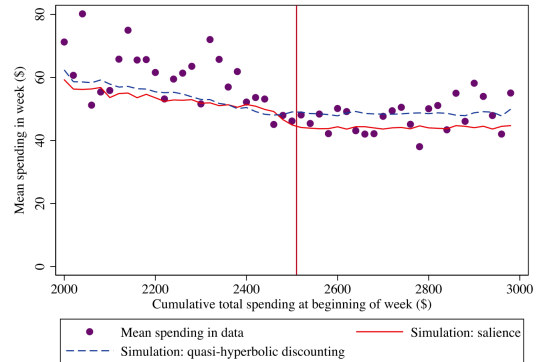


FIGURE 5
Fit of quasi-hyperbolic discounting and price salience models near coverage gap

(b) Figure 5: model fit

6.6 Tables 6 and 7: counterfactuals and policy design

Read:

- Table 6 compares the estimated baseline to a benchmark model with no behavioral hazard and to a no-insurance case.
- Moving from the baseline to the benchmark reduces enrollee spending from \$15.82 to \$10.91 per week and total spending from \$41.20 to \$35.08.
- Removing drug insurance entirely lowers total spending further to \$26.61, but also lowers welfare sharply to 0.63.

- Table 7 shows that removing the doughnut hole increases total spending from \$41.26 to \$45.32 and raises insurer spending from \$25.55 to \$32.49.
- A constant-spending no-gap policy raises welfare to 1.31 while holding insurer spending at \$25.55, but it changes incentives through a 37% coinsurance rate.
- Generic-only gap coverage raises welfare to 1.29 with insurer spending of \$27.23, a smaller fiscal cost than full gap elimination.

Economic message: Behavioral hazard and insurance both matter. The gap reduces insurer spending, but it also creates welfare losses because consumers respond to a salient current price in a dynamically distorted way.

Additional interpretation: The policy lesson is nuanced. Eliminating the gap improves consumer welfare but is expensive for insurers. Generic-only coverage or linear coinsurance may deliver some welfare gains at lower fiscal cost.

TABLE 6
Relative impact of behavioural hazard and drug insurance

Statistic (per week)	Baseline: $\sigma = 0$ future not salient Case 1	Benchmark model (no behavioural hazard) Case 2	No drug insurance Case 3
Number of Rx's	0.59	0.57	0.42
Number of branded Rx's	0.16	0.13	0.11
Number of generic Rx's	0.35	0.38	0.25
Expensive Rx's	0.08	0.06	0.05
Medium Rx's	0.17	0.15	0.13
Inexpensive Rx's	0.33	0.36	0.25
Enrollee spending (\$)	15.82	10.91	26.61
Insurer spending (\$)	25.39	24.17	0.00
Total spending (\$)	41.20	35.08	26.61
Enrollee welfare	1.21	1.27	0.63

Notes: Simulations use estimated parameters from Table 5 column 3. Inexpensive Rx's are less than \$50 and expensive ones are \$150 or more. Simulations are performed for 52 weeks starting enrollees at \$0 in expenditures and use a mix of the estimation sample and other enrollees in same plans so that 33% reach the doughnut hole in the base case. Geometric discounting case uses an annualized discount factor of 95%.

(a) Table 6: behavioral hazard

TABLE 7
Impact of filling the doughnut hole

Statistic (per week)	Baseline: $\sigma = 0$ future not salient Policy 1	No doughnut hole Policy 2	No doughnut hole with constant insurer spending Policy 3	No doughnut hole for generics only Policy 4
Number of Rx's	0.59	0.63	0.62	0.61
Number of branded Rx's	0.16	0.18	0.17	0.16
Number of generic Rx's	0.35	0.37	0.37	0.39
Expensive Rx's	0.08	0.10	0.08	0.08
Medium Rx's	0.17	0.18	0.16	0.17
Inexpensive Rx's	0.33	0.35	0.38	0.36
Enrollee spending (\$)	15.71	12.83	14.88	14.25
Insurer spending (\$)	25.55	32.49	25.55	27.23
Total spending (\$)	41.26	45.32	40.43	41.48
Enrollee welfare	1.21	1.35	1.31	1.29
Coinsurance rate	From data	From data	37%	From data

Notes: Simulations use estimated parameters from Table 5 column 3. Inexpensive Rx's are less than \$50 and expensive ones are \$150 or more. Simulations are performed for 52 weeks starting enrollees at \$0 in expenditures and use a mix of the estimation sample and other enrollees in same plans so that 33% reach the doughnut hole in the base case.

(b) Table 7: filling the doughnut hole

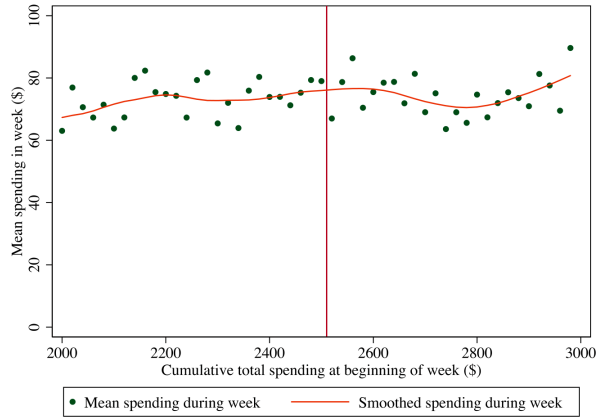
6.7 Appendix figures: falsification and full-sample spending shape

Read:

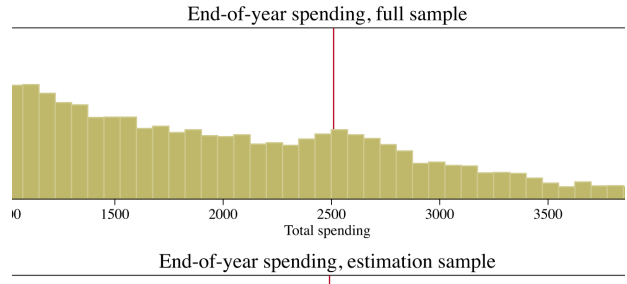
- Figure A2 repeats the gap-style analysis for Plan F, whose actual gap starts at \$4,000 rather than \$2,510.
- There is no analogous sharp drop at \$2,510 for Plan F, supporting the claim that the main result is tied to the true contract threshold.
- Figure A3 shows end-of-year spending distributions. The estimation sample is selected from enrollees with a high probability of reaching the gap.

Economic message: The falsification figure reduces the concern that \$2,510 is merely a coincidental health-spending point. The response appears where the insurance contract changes.

Additional interpretation: The appendix figures help convert the reduced-form discontinuity from a descriptive graph into a stronger behavioral test of salience and dynamic incentives.



(a) Figure A2: Plan F falsification



(b) Figure A3: spending distributions

7 Short summary

- The paper studies how Medicare Part D enrollees respond to the nonlinear doughnut-hole price schedule.
- The benchmark model predicts no sharp spending drop at the gap for enrollees who expect to enter the gap anyway.
- The data show the opposite: spending is flat before the gap and falls sharply after gap entry.
- The drop is large for total spending, prescription counts, branded drugs, and expensive drugs.
- A low geometric discount factor does not fit well because it predicts a pre-gap spending decline.
- Structural estimates show current-price sensitivity but essentially no salience of future prices.
- The price-salience model fits better than quasi-hyperbolic discounting.
- Behavioral hazard is quantitatively important: full future-price salience would lower enrollee spending by 31% and total spending by 15%.
- Removing the doughnut hole raises welfare but also raises insurer spending substantially.
- More targeted redesigns, such as generic-only gap coverage or linear coinsurance, can mitigate fiscal costs while reducing behavioral distortions.

8 Conclusion

8.1 Core conclusion

Dalton, Gowrisankaran, and Town show that consumers facing a complex nonlinear health-insurance contract do not behave like fully forward-looking optimizers. In the Medicare Part D doughnut hole, enrollees who should expect to enter the gap do not reduce spending before the gap. Instead, they cut spending sharply only after the high out-of-pocket price becomes current. This timing is consistent with limited salience of future prices.

8.2 Incremental contribution

The paper’s incremental contribution is to connect three literatures that are often separate. First, it contributes to health-insurance moral hazard by showing that the relevant distortion is not only current price sensitivity, but also behavioral hazard from misunderstanding or ignoring future prices. Second, it contributes to behavioral industrial organization by building and estimating dynamic discrete-choice models with alternative behavioral frictions. Third, it contributes to public finance by evaluating contract redesign in a setting where consumer mistakes and insurance protection interact.

Relative to studies of Part D plan choice, the paper studies behavior *after* plan choice and therefore asks a different question: conditional on having a plan, can consumers optimize within a complicated dynamic incentive system? Relative to reduced-form studies of cost sharing, it uses the shape of behavior around the gap to distinguish full dynamic optimization, low discounting, present bias, and low salience. Relative to purely theoretical work on nonlinear pricing, it provides a structural estimate of how salient future nonlinear prices are.

8.3 Interpretation of the empirical evidence

The reduced-form and structural evidence tell a coherent story. The flat spending path before the gap indicates that consumers are not steadily adjusting to the future price. The sharp drop at the gap indicates that consumers are responsive once the price is immediate. The compositional changes away from branded and expensive prescriptions show that the response is economically meaningful and not just noise in timing. The Plan F falsification helps show that the response is tied to the true insurance threshold.

The structural estimates reinforce this interpretation. Consumers have significant price elasticities, so the issue is not a lack of price response. Instead, the estimated salience parameter says that future prices are not incorporated into current decisions. This is why the salience model fits the discontinuity better than the quasi-hyperbolic model.

8.4 Policy implications

The findings imply that nonlinear insurance contracts can have unintended behavioral consequences. A doughnut hole can reduce insurer spending, but it can also induce abrupt reductions in drug use exactly when consumers face high prices. If some of these drugs have health value, this may create downstream medical costs or health losses.

The paper also shows that simple elimination of the doughnut hole is expensive. Full gap elimination increases insurer spending by 27%, while generic-only gap coverage increases insurer spending less and still raises welfare. This suggests that policy design should compare not just generosity, but also salience, transparency, and the composition of drug use.

8.5 Implications for contract design

A central lesson is that complexity matters. A nonlinear contract can be optimal under rational expectations but perform poorly if consumers do not understand or internalize future prices. Contract designers should therefore ask whether consumers can realistically calculate the dynamic consequences of current choices. Linear coinsurance, simplified coverage tiers, clearer real-time price information, or targeted coverage for high-value medications may improve behavior even without increasing total generosity.

8.6 Limitations

The paper does not model all medical dynamics of treatment, substitution between drugs and non-drug care, physician agency, or long-run health outcomes. The welfare calculations are therefore informative but not

definitive. The results are also based on a specific set of employer-sponsored plans in 2008. Extrapolation to other populations or modern Part D designs should be done cautiously.

8.7 Research agenda

Future work could link drug-use changes to hospitalization, mortality, and long-run chronic disease control; test whether real-time price reminders improve salience; compare nonlinear and linear benefit designs experimentally; and study whether physicians, pharmacists, or plan interfaces can act as intermediaries that make future prices more salient.

8.8 Takeaway

The enduring takeaway is that consumers can be price sensitive yet dynamically inattentive. In complex insurance contracts, the salient current price may matter more than the actuarially relevant future price. That distinction is central for designing health insurance, evaluating moral hazard, and measuring welfare in markets with nonlinear pricing.